

Supplement (v2) to: Does a one-pool water-balance model improve forecasts of Edwards Aquifer head? A scored test at J-17

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Edwards Aquifer head? A scored test at J-17”

Companion to carrier of record paperE3_edwards_forecast_ladder_v24_humanized (stamp se24-c1375ed; values below unchanged from the frozen layers they cite). Section S1 carries the independent replication of the uncertainty layer of Section 5.3.1; Section S2 carries the truncated-detail climate fixed-window record of Section 5.4. All values are archived with the analysis code and re-compute from the registered per-origin forecast files.

S1 Independent replication of the uncertainty layer

An independent replication of this layer is registered with the repository: the archived script `campaign_e3_dm_uncertainty.py`, a second Diebold–Mariano + moving-block-bootstrap implementation — HAC lag $h - 1$ with unweighted truncation and population-variance scaling, block length h , 20,000 replications, seed 0, deterministic (byte-identical on re-execution) — reading the same registered per-origin forecast file. Its verified output (archived alongside it) reproduces every load-bearing reading: the M1 retention margin DM $z = -0.86$ with block CI $[-1.22, +0.56]$ ft (bootstrap $p = 0.38$); the M2m edge over persistence ($p = 0.001$); the $h = 5$ climatology win ($p = 0.035$). It also registers two rows this table does not carry — the $h = 1$ climatology loss (+2.94 ft against persistence, $p = 0.046$) and the M2 $h = 5$ loss (+12.4 ft, $p < 0.001$) — both consistent with the readings above. The two implementations use different seeds, block lengths, HAC weighting, and variance conventions, and agree on every conclusion: the 0.39-ft AR(1) retention margin is statistically indistinguishable from zero, the M2m edge is the only one-year margin that separates from noise, and the five-year climatology win survives.

S2 Climate-informed modules on the remaining fixed windows

The remaining fixed windows complete the same record: drawdown 1951–56, M2_Renso 24.30, M2_Rprecip 28.67, M2_combo 24.70, precipitation oracle 18.16 ft; pre-permit wet 1991–95, M2_Rprecip 22.03, M2_Renso 23.03, M2_Rar 23.94, M2_combo 25.71, oracle 10.98 ft; critical-period era 2015–23, M2_Rprecip 14.52, M2_Rar 14.67, M2_Renso 16.01, M2_combo 16.57, oracle 9.75 ft — on this one window the two climate-informed modules M2_Rprecip and M2_Rar edge past M1 by about one foot (14.52 and 14.67 versus 15.62 ft). On the recharge target itself the fixed-window scores are an order of magnitude coarser on every window (climate modules 199–937; precipitation oracle 80.7–487, against the 556×10^3 acre-ft climatology scale of the rolling record), so the marginal head advantage is a window-specific result, not a recharge forecast, and it does not enter the abstract.

S3 Reproducibility package

All values in the carrier and in S1–S2 recompute from the registered scripts below, archived at <https://github.com/MIKEAA2020/general-sustainability>; checksums (md5, first 8 hex) pin the release state. Execution is deterministic throughout — a fresh environment regenerated every archived result file byte for byte. Two independent DM/bootstrap implementations (10,000 and 20,000 replications, different seeds, block lengths, HAC weighting, and variance conventions) agree on every load-bearing conclusion; both are registered, and the nClimDiv precipitation workbook is the sole raw file not redistributed (provenance URL archived in the sources index; scoring from the registered analysis panel does not require it).

Script (repo path)	Role in the paper	md5	
wave_e_edwards/src/build_panel.py	Builds the registered twenty-column analysis panel (dataset of record)	2cfd1649	results analysis panel
wave_e_edwards/src/build_climate.py	Builds the climate (Niño 3.4, precipitation) panel columns	356af249	climate panel
wave_e_edwards/src/run_ladder.py	J-17 model ladder scoring, persistence and oracle diagnostics	e01f29cd	results rolling_forecasts.csv, window tables
wave_e_edwards/src/e3_audit_uncertainty.py	Unity post-freeze uncertainty layer: DM + Newey-West HAC + moving-block bootstrap, 10,000 replications, seeded, deterministic	074ff015	wave_e_edwards/results/e3_audit_uncertainty.json
batch 7 (audits of agent arena 1 paper rewrites)/campaign_e3_dm_uncertainty.py	Independent replication of the same layer: NB = 20,000, seed 0, byte-identical re-execution	30963493	results/ e3_dm_uncertainty.csv
arena_agent_1/other documents/rerun_campaigns/campaign_e3_pumpage_scenarios.py	Pumpage counterfactual layer of Section 5.6	c3fe4fe3	scenario table outputs
wave_e_edwards/src/make_figures.py	Figures 1–5 of the main text	57c17e29	figs/fig1_series... fig5_fibre